

With the additional visualizations from layers 5 and 11 (head 0) in the new PDFs, I can refine my analysis. These show clearer evolution in attention patterns across layers, confirming and expanding on my initial generalizations. I've incorporated specific observations from the screenshots, such as line strengths (indicating attention scores) and QKV interactions (via described hover details, though static—e.g., higher dot products for semantically key pairs in deeper layers).

1. What similarities and differences do you notice in the visualizations between the examples in part (a)? Explore the queries, keys, and values to identify any interesting patterns associated with the attention mechanism.

Similarities: In all three examples ("The dog ran", "The dog sitting in the car", and "Jaime Salazar is running for election. Despite his party affiliation, Mr."), the attention remains autoregressive and causal, with tokens only attending to prior ones (no future connections, forming a lower-triangular pattern). Stronger lines (high scores ~1) typically link semantically or syntactically core elements, while faint lines (~0) ignore irrelevant context. QKV patterns show queries from later tokens aligning with keys from nouns/verbs for value aggregation—e.g., across examples, last tokens' queries yield high dot products with subject keys, building context incrementally. This highlights attention's role in sequential dependency capture.

Differences: The simple sentence ("The dog ran") has sparse, local attention—e.g., "ran" strongly attends to "dog" (subject-verb), with faint to "The"; QKV shows low-variance dot products focused on immediate syntax. The complex phrase ("The dog sitting in the car") extends to mid-range—e.g., "car" attends to "the" (strongest), "in", "sitting" (modifiers), weakly to "dog"/"The"; queries emphasize positional keys for prepositions. The long-context example ("Jaime Salazar... Mr.") reveals long-range resolution—e.g., "Mr." attends strongly to "Salazar"/"Jaime" (coreference), "his"/"party" (pronoun/anaphora); QKV dot products peak for distant proper nouns, with values fusing entity history. **Interesting patterns:** In complex cases, queries for pronouns/periods show blue-positive weights to antecedents, while simpler ones have orange-negative for non-adjacent—demonstrating attention's adaptability to context length.

2. How does attention differ between the different layers of the GPT model? Do you notice that the tokens are attending to different tokens as we go through the layers of the network?

Attention shifts from local/syntactic in shallow layers (e.g., layer 0) to global/semantic in deeper ones (layers 5, 11), a hallmark of transformer hierarchy. In layer 0, patterns are adjacent-focused—e.g., in "The dog ran", "ran" attends mostly to "dog" (short lines); in "Jaime Salazar...", "Mr." links weakly to nearby "affiliation,", ignoring distant. QKV: Dot products are position-dominated, with values aggregating basic order.

By layer 5, attention broadens mid-range—e.g., "ran" attends strongly to "dog" but more to "The"; in "The dog sitting in the car", "car" spreads to "the"/"in"/"sitting" (modifiers emphasized); in the election example, "Mr." strengthens to "Salazar"/"Jaime" (coreference emerging). Tokens shift: Early focus on neighbors (e.g., articles-nouns); now mid-context (pronouns to subjects). QKV: Queries abstract slightly, with higher variances in dot products for relations.

In layer 11, it's highly diffuse/global—e.g., "ran" attends nearly equally across all prior; "car" distributes evenly to all tokens; "Mr." broadly weights name, "his", "party", "election" (thematic integration). Yes, tokens attend differently: Shallow layers to positional/syntactic (e.g., "his" to "Despite"); deep to semantic/distant (e.g., "his" to "Jaime"). QKV: Deeper queries prioritize content similarity, producing values for high-level representations—enabling GPT's coherent generation.

Next, let's focus on part b), which includes questions 3, 4, 5, 6, and 7. Before you provide your answers, are there any specific layers, heads, or attentions you would like to view a PDF for?

Yes, to accurately analyze questions 3, 5, and 6 (which involve examining patterns across different layers in BERT, differences in embeddings for the ambiguous sentences, and bidirectionality), I'd like to see visualizations for the BERT sections in part (b). The current PDFs primarily capture GPT at varying layers, with BERT fixed at layer 2, head 0.

Please provide notebook PDFs showing:

- For the main BERT example ("I wanted to have a party [SEP] I bought a cake for it."): Layer 0, Head 0; Layer 5, Head 0; Layer 11, Head 0.

- For the ambiguous sentences ("T was going to play at the park." and "I was going to a play in the park"): Layer 0, Head 0; Layer 5, Head 0; Layer 11, Head 0 for each.
- For the reversed example ("party a have to wanted I [SEP] I bought a cake for it"): Layer 0, Head 0; Layer 5, Head 0; Layer 11, Head 0.

If separate PDFs per combination are too many, combined ones with those dropdown selections would work. This will help observe layer-specific patterns in BERT's bidirectional attention.

Here are the notebooks where I have assigned all visualizations to Layer 0, Head 0; Layer 5, Head 0; and Layer 11, Head 0 respectively. Please compare between these as necessary to answer the questions. With this additional information, please analyze the visualizations and answer questions 3-7 for part b).

PDF Visualizing_BERT_layer_11_head0.pdf

PDF Visualizing_BERT_layer0_head0.pdf

PDF Visualizing_BERT_layer5_head0.pdf

Revised Answers to Questions 3-7

Based on the detailed visualizations from layers 0, 5, and 11 (head 0) in the provided PDFs, I've refined my analysis. The screenshots capture the attention maps for the key examples in part (b): the "party/cake" pronoun resolution, the ambiguous "play" sentences ("I was going to play at the park."—note the typo as "T" in code, but contextually "I"), and the reversed inputs. Patterns are observed from line strengths (attention scores), directions (bidirectional), and implied QKV (via hover descriptions, though static—e.g., blue positives for relevant pairs).

3. Look at different layers of the BERT model in the visualizations of part (b) and identify different patterns associated with the attention mechanism. Explore the queries, keys, and values to further inform your answer. For instance, do you notice that any particular type of tokens are attended to at a given timestep?

In the "I wanted to have a party [SEP] I bought a cake for it." example, layer 0 shows local, syntactic patterns: Short, strong lines between adjacent tokens (e.g., "have" to "to", "cake" to "a"), with faint global connections; [CLS] attends narrowly to start, [SEP] ignored. Mid-layer 5 broadens to mid-range dependencies: "it" strongly attends to "party" across [SEP] (pronoun resolution), with denser lines; queries for pronouns yield high dot products with noun keys. Layer 11 is global/semantic: Broad, even distribution (e.g., "it" attends widely to theme words like "party", "cake"); values aggregate holistic context, [CLS] summarizes all.

For ambiguous sentences ("I was going to play at the park" vs. "I was going to a play in the park"), layer 0 is similar/positional. Layer 5 diverges: Verb "play" attends to "going"/"at" (action flow); noun "play" to "a"/"in" (object modifiers). Layer 11 amplifies semantics: Verb links to "park" as activity; noun to "park" as location.

In reversed ("party a have to wanted I [SEP] I bought a cake for it"), patterns hold bidirectionally—layer 0 local, 5 resolves "it" backward to "party", 11 thematic.

Particular tokens: Pronouns ("it") attend to antecedents (nouns) in mid/late layers; [CLS] to all for summary; [SEP] low attention as boundary. QKV: Queries in deeper layers favor semantic keys (high positives for relations), early positional.

4. Do you spot any differences between how attention works in GPT vs. BERT? Think about how the model architectures are different.

From prior GPT (unidirectional, causal) vs. BERT visuals: GPT has triangular patterns—tokens attend only leftward (e.g., no future in "Jaime... Mr.", "Mr." to past names). BERT is fully connected bidirectionally (lines everywhere, e.g., "it" backward to "party"). In GPT, layer 0 local, 5 mid, 11 global but masked; BERT unmasked, enabling context from both sides (e.g., reversed still resolves). Architecturally, GPT's decoder masks for autoregression/generation; BERT's encoder with MLM/NSP pre-training allows bidirectional encoding for understanding. QKV in BERT computes over full sequence (symmetric); GPT masked (asymmetric values).

5. Look through the different layers of the two BERT networks above associated with sentence a and sentence b, and take a look at the queries, keys, and values associated with the different tokens. Do you notice any differences in the embeddings learned for the two sentences that are essentially identical in structure but different in meaning?

Yes, embeddings differ contextually despite structure. In "I was going to play at the park" (verb "play"), layer 0 similar/positional. Layer 5: "play" attends strongly to "going" (sequence), "at" (preposition)—queries emphasize dynamic keys. Layer 11: Broad to "park" as activity venue, values fuse motion.

In "I was going to a play in the park" (noun "play"), layer 5: "play" to "a" (article), "in" (location)—queries weight nominal keys. Layer 11: Stronger to "park" as event site, values static/object-oriented. QKV: Verb "play" dot products high with temporal/verb keys (blue positives); noun with determiner/location (differing vectors capture polysemy via context).

6. Do you notice BERT's bidirectionality in play?

Yes, evident in reversed examples. In "party a have to wanted I [SEP] I bought a cake for it", layer 5/11 still connect "it" strongly to "party" (lines cross backward), resolving reference despite order. In "I bought a cake for it [SEP] I went to a party...", forward links persist. Bidirectionality ignores linearity, fusing left/right context—unlike GPT's one-way.

7. Do you think pre-training the BERT helped it learn better representations?

Yes, pre-training (MLM/NSP) enables structured patterns: Meaningful, sparse connections (e.g., pronoun-antecedent) vs. untrained's uniform/faint (part d). It teaches contextual embeddings, improving resolution/bidirectionality for downstream—untrained lacks peaks in QKV dot products.

Could you provide a more concise answer to question 3? Is there any particular token that receives a large amount of attention from other tokens at a certain layer?

Concise Answer to Question 3

Attention patterns in BERT shift across layers: Layer 0 focuses on local, syntactic connections (e.g., adjacent tokens like "have" to "to"). Layer 5 emphasizes mid-range dependencies (e.g., "it" strongly to "party" for pronoun resolution). Layer 11 is global and semantic, with broad, diffuse attention across tokens. Particular tokens: Pronouns attend to antecedents in mid/layer layers; [CLS] aggregates from all for summary; [SEP] receives low attention as a boundary. QKV analysis reveals early positional dot products, deeper semantic alignments.

Yes, the [CLS] token receives a large amount of attention from other tokens in layer 11, as seen in the broad converging lines, enabling it to pool sentence-level information.

Now, let's focus on part c). In answering questions 8 and 9, what are different features being learned and focused on between different attention heads?

Answers to Questions 8 and 9

8. Do you notice different features being learned throughout the different attention heads of BERT? Why do you think this might be?

Yes, the visualization on page 6 shows clear specialization across the 12 heads per layer. For example, in early layers (0–3), some heads (e.g., 0–2) show mostly diagonal or short-range lines, indicating local focus, while others (e.g., 9–11) have vertical patterns converging on special tokens. In mid-layers (4–7), heads diverge more: some maintain sparse connections, others form blocks within sentences A or B. Later layers (8–11) have denser, long-range patterns in certain heads (e.g., 3–5), with others diffuse. This specialization occurs because multi-head attention linearly projects the input into independent subspaces, allowing each head to capture unique relational aspects (e.g., syntax vs. semantics) in parallel, improving overall model capacity and preventing a single representation from dominating.

9. Can you identify any of the different features that the different attention heads are focusing on?

From the grid visualization (layers rows, heads columns), patterns suggest:

- **Syntactic features:** Heads 0–2 (across layers) show diagonal/adjacent lines, focusing on local dependencies (e.g., "to" to "have", "a" to "party" as preposition-object).
- **Coreference/pronoun resolution:** Heads 3–5 in mid-layers have lines crossing [SEP], linking "I" in sentence A to "I" in B, or "party" to "dinner" as related events.
- **Semantic/thematic features:** Heads 6–8 display longer, thematic connections (e.g., "Thanksgiving" to "dinner" as compound noun, "party" to "dinner" for event similarity).
- **Positional/special token features:** Heads 9–11 often converge vertically on [CLS] (aggregation) or [SEP] (sentence boundaries), with broad or uniform patterns; some ignore content for positional cues. Early heads are more positional/syntactic, later more abstract/semantic.